# Adaptive Scale-Aware Semantic Memory Network for Remote Sensing Image Captioning

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***Abstract*—Remote sensing image captioning between visual images and natural language remains a long-standing challenge in the remote sensing community. Due to the wide coverage and large amount of information in remote sensing images, existing methods struggle to effectively utilize the relevant semantic information about objects and their attributes at different scales across samples to generate descriptions. To address these issues, the article proposes a novel adaptive scale-aware semantic memory network (ASSMN) for remote sensing image captioning. First, to fully extract the semantic information in remote sensing images, multilevel feature enhancement is constructed to improve the feature representation extracted from the contrastive language-image pretraining (CLIP) pretraining model. Subsequently, a scale-aware attention aggregator (SAA) is introduced to further integrate the enhanced multiscale image features into the high-level semantics of remote sensing images. Then, to fully exploit the semantic information of the joint observed samples, a semantic memory reinforcement (SMR) is designed to strengthen the semantic representation of the current scene through the relevant semantics obtained from other training samples. Finally, a captioning decoder is performed to generate a comprehensive scene caption with accurate objects and attributes. In the experiments, the performance of the proposed ASSMN on three remote sensing image captioning datasets is evaluated and compared with well-designed baselines and state-of-the-art methods, and the superiority of the proposed method is demonstrated by ablating the role of each proposed component. The code will be available at https://github.com/zcsisiyao/ASSMN**

***Index Terms*—Contrastive language image pretrain, image captioning, prototype memory, remote sensing.**

## I. Introduction

REMOTE sensing image captioning is a multimodal technology that integrates computer vision with natural language processing to generate textual descriptions for earth observation [1]. This complex task is beyond the capabilities of traditional remote sensing tasks [2], [3], [4] and requires text-based human-like interpretation of spatial and scene remote sensing image information [5], [6]. Remote sensing image captioning plays a crucial role in a wide range of applications such as natural disaster management [7], smart city development [8], and military intelligence collection [9]. By expanding the boundaries of the semantic understanding of remote sensing data, remote sensing image captioning is advancing these fields.

In the field of remote sensing image captioning, similar to natural image captioning, techniques based on the encoder–decoder paradigm [10], [11], [12], [13], [14] have become the mainstream trend due to their concise structure, flexible sentence length, and natural description. Early approaches utilize a convolutional neural network (CNN) [15] or vision transformer (ViT) [16] as a visual encoder to capture image features, and then deep sequence models such as recurrent neural network (RNN) [17] or Transformer [18] is used as language decoders to convert image features into natural language. However, different from natural images, remote sensing images typically encompasses objects of varying scales [19] and categories [20] due to its top-down imaging perspective. Furthermore, the multitude of objects forms complex spatial attributes [21]. These characteristics complicate the integration of features across different scales within remote sensing images and the capture of relationships between them. Therefore, for remote sensing image captioning, effectively abstracting and characterizing the remote sensing images while identifying objects and their associated attributes is crucial for generating accurate semantic descriptions.

To alleviate these issues, current remote sensing image captioning approaches can be summarized into two types. The first one, shown in Fig. 1(a), is single visual encoder-based methods [19], [22], [23]. They utilize visual features extracted at different scales from a visual backbone network, then aggregate these features using various types of attention mechanisms or spatial pooling methods. Finally, an image description is generated through a text decoder. These image description methods based on single-visual encoders can effectively capture objects of varying scales and categories within remote sensing images, while offering high efficiency and versatility. However, due to the semantic gap between vision and language features, and the lack of sufficient interaction between image features at different scales. These may lead to the generation of partial semantic descriptions, such as quantitative descriptions of similar targets at different scales.

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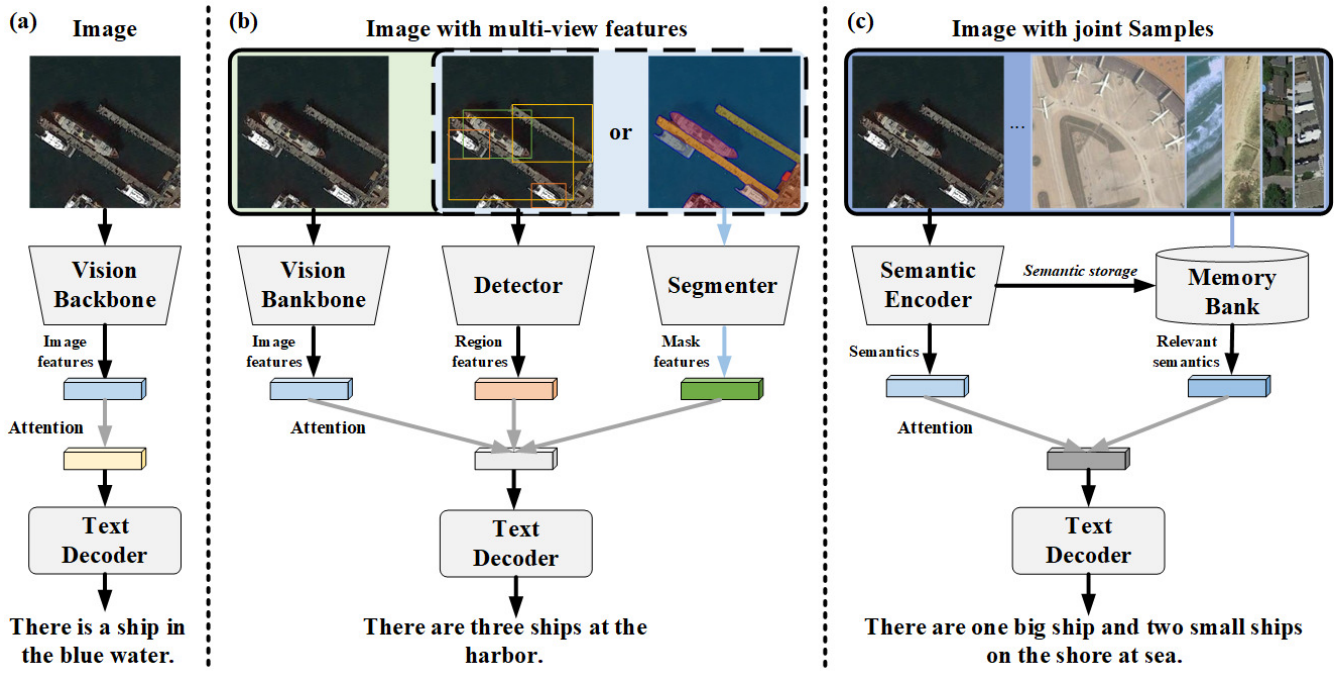



Fig. 1. Comparisons between the current paradigm of captioning model (a) single visual encoder-based framework, (b) hybrid visual encoder-based framework, and (c) our adaptive scale-aware semantic memory framework for remote sensing image captioning.

The other type, as shown in Fig. 1(b), is hybrid visual encoder-based approaches [7], [24], [25], which are based on a single visual encoder that jointly utilizes image features of different views captured by a visual feature extractor, such as region features encoded by an object detector [13] or mask features encoded by a semantic segmenter [33]. The caption is then generated by attention fusion of different views features. The addition of supplementary view features can extend the understanding of the model for the current remote sensing scene with respect to the object and its attributes. However, the language description is generated only based on different view features for a single image are limited by the performance of the special view extractor [26], [27], which tends to produce rich but ambiguous descriptions.

Therefore, for remote sensing image captioning, effectively integrating both types of image captioning methods while efficiently combining multiscale feature aggregation with supplementary additional perspective features is crucial for generating appropriate objects and attributes descriptions. Recently, methods leveraging semantic information from joint observation samples as supplementary additional perspectives have been explored in natural image captioning [32], proving to be efficient and practical approaches. However, such methods remain underexplored in the context of remote sensing images. Based on the above ideas, the article proposes a solution that combines deep semantic encoding and global memory reinforcement for remote sensing image captioning, as shown in Fig. 1(c). First, to better capture objects and their attributes at different scales in remote sensing images, multilevel feature enhancement (MFE) is proposed to enhance the feature representation capability by contrastive language-image pre-training (CLIP) [29]. Then, to migrate the relationship between different scales of information to further reduce the semantic gap, a scale-aware attention aggregator (SAA) is introduced for integrating features at different scales to generate a semantic representation of the current image. Finally, accessing past training samples at generation time is considered to be a powerful source of information that can improve the quality of the description [32]. Semantic memory reinforcement (SMR) is designed to process the attention acquired in other training samples. The semantic memory models the distribution of past samples by defining prototype vectors. Relevant semantic information from joint observations of historical input samples is fully considered for the description of the current scene. The main contributions of this work can be summarized as follows.

1) A novel global semantic memory-enhanced remote sensing image captioning framework is proposed, which focuses on integrating the semantic information from remote sensing images at different scales. adaptive scale-aware semantic memory network (ASSMN) efficiently utilizes relevant information from samples to generate accurate and complete captions.
2) To obtain complete semantic information of remote sensing images, MFE is proposed to enhance the CLIP feature representation capability at different scales, and then the SAA module is introduced to further aggregate the features at different scales and generate a semantic representation of the image.
3) The SMR is designed to enhance the interaction between the current image information and other image information so that relevant semantic information is activated to enhance the understanding of complex scenes.

The remainder of this article is organized as follows. Section II discusses related work on natural image captioning with memory networks and remote sensing image captioning.

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Section III details the proposed ASSMN. Section IV introduces remote sensing image captioning datasets and analyzes the experimental results of the proposed method and ablation study. Section V provides a conclusion of this approach.

## II. Related Work

In this section, the relevant work on natural image captioning and memory networks will be introduced first, followed by the related work on remote sensing image captioning.

### A. Natural Image Captioning With Memory Network

Image captioning involves generating textual descriptions for images by combining vision and language models. The development of image captioning can be delineated into two distinct phases: the traditional machine learning stage and the advent of deep learning approaches [34], [35]. In the earlier machine learning stage, template-based [36] and retrieval-based [37] methods are commonly employed. However, with the emergence of deep learning, inspired by the success of encoder–decoder frameworks in the field of machine translation [38], [39], the CNN–RNN framework is extensively used in image captioning and achieves great success [40], [41], [42]. For example, Xu et al. [43] introduced an embedding model of both image and text to build up a multimodal sentence production model, utilizing a CNN as the encoder and a long short-term memory (LSTM) as the decoder, and [44] further proposed a reinforcement learning framework. In a later work, [45] designed a fine-grained region feature extractor using an R-CNN object detector [46], enabling the generation of region-level captions for the given image.

Nowadays, attention and transformer-based architectures [18], [47] are often employed both in the visual encoding stage, usually applied to refine features from a CNN or ViT, and as language models [48], [49]. Other solutions [50], [51] utilize self-attention to effectively combine visual features from multiple encoders, such as typical CNNs and object detectors, and in some cases, fine-tune the visual encoder to improve the final performance. A common strategy is that of employing visual features extracted from large-scale cross-modal architectures [52], [53] like CLIP. As done in [54], [55], and [56], these multimodal architectures can also enable the enrichment of predicted textual sentences through retrieval components that can be added to the captioning model. Memory banks have been used in different ways in encoder–decoder architectures [57]. Khandelwal et al. [58] structured the language generation memory as a large table consisting of key and token pairs, while Wu et al. [59] propose a memorizing transformer architecture in which they retrieve activations produced over long documents. Recently, SAM-2 [60] applied a memory bank to video instance segmentation. The memory bank in SAM-2 stores instance-level features from previous frames and uses them as conditions for segmenting subsequent frames, ensuring segmentation consistency for the same object. In vision and language, learnable external memories have been successfully employed for image captioning [48], visual relationship recognition [61], and story generation [62]. The external memory has also been used for multimodal data generation [63], [64], for circumventing the mode collapse issue and preserving detailed data structure. However, despite the progress achieved in natural images using memory-enhanced vision-language representations, their performance is restricted when applied directly to complex remote sensing images. In this case, we can refer to these methods to design corresponding algorithms for remote sensing image captioning.

### B. Remote Sensing Image Captioning

In the field of remote sensing images, captioning algorithms based on encoder–decoder architectures are employed, and they are capable of adapting to differences in data sources, feature translations, and semantic representations. Qu et al. [65] used a CNN-RNN framework and demonstrated on their proposed datasets, UCM-captions and Sydney-captions, that the framework applied to natural images also achieves good performance when processing remote sensing images. Furthermore, Zhang et al. [66] proposed a global visual feature-guided attention mechanism to filter out redundant feature components in the fused local and global features through an attention gate. However, these methods do not express multiscale information about remote sensing images, which are crucial for the remote sensing image captioning task. To effectively utilize the features of different scales, Huang et al. [22] extracted multiscale features using VGG and employed a transformer encoder for adaptive feature aggregation, and finally generated descriptions through a visual-text alignment LSTM. Liu et al. [23] fused visual features from different high-level layers of ResNet-50 to present multiscale information, followed by an aggregation of these features using LSTM within a transformer architecture to utilize the features effectively. In contrast, Zhang et al. [67] introduced an attribute-based attention mechanism that integrates low-level features with high-level attributes from the VGG-16 model, subsequently employing LSTM for sentence generation. Wu et al. [14] proposed a cross-modal reason dual transformer approach that leverages the Swin Transformer to extract multiscale features for sentence generation. Considering the noise of features from different layers, Huang et al. [68] proposed a denoising-based multiscale feature fusion mechanism to improve the ability of multiscale feature representations. Besides, grid-based visual features are introduced to improve remote sensing image captioning performance by Zhuang et al. [69]. ViT [16] is brought into remote sensing image captioning by Ren et al. [70], allowing local patch features to explore the rich information of remote sensing images. Yang et al. [19] proposed to aggregate different layers of extracted features based on CNN. Considering the discrepancy between multimodal features, cross-modal feature alignment loss is further used to align visual features and text features.

To better align image and text features without adding additional loss functions, it becomes a compromise to directly apply image encoders that have been pretrained with vision and language alignment [30]. For example, Lin et al. [28] utilized CLIP visual grid features and a stochastic masking strategy for remote sensing image captioning. Moreover, Meng et al. [71] utilized a hybrid image encoder composed of CLIP and CNN for richer image representation and further focused on different contextual information by a global

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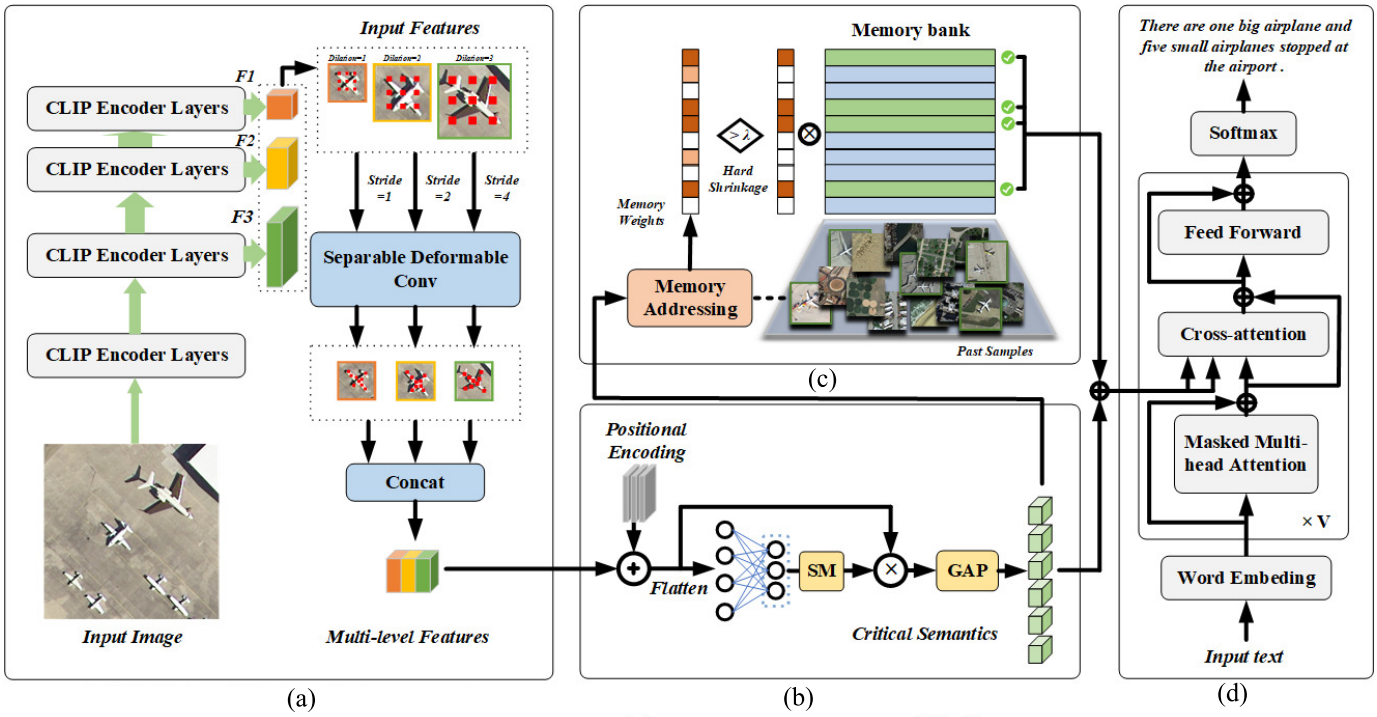



Fig. 2. Overview of ASSMN, consisting of (a) MFE, (b) SAA, (c) SMR, and (d) captioning decoder. Each connection in the flowchart signifies the flow of information among the components, with solid arrows indicating the primary processing path from the input image to output caption. ⊕, ⊗, SM, and GAP represent matrix addition, matrix multiplication, SoftMax, and global average pooling, respectively.

grouping attention mechanism. Kandala et al. [72] employed a transformer-based network for remote sensing image captioning. In particular, to handle limited training data, an auxiliary decoder is used for multilabel scene classification that exploits the conceptual similarity between the encoder and the image captions and the ability to highlight semantic classes, assisting the encoder during the training process. Meng et al. [7] proposed a prior knowledge-enhanced attention mechanism for remote sensing image captioning, where target information and scene information are represented by faster-RCNN and CNN extractors and finally integrated by a transformer.

Despite significant advances in multiscale and multiview methods, they usually fail to adequately account for the interactions between visual and textual features at different scales in all samples at the generation stage, leading to inaccurate captioning. Inspired by [32] and [48], we propose an adaptive scale perception framework based on memory enhancement to alleviate these problems. MFE is constructed to enhance multilevel feature representations through different ratios of dilation-deformable convolutions. Furthermore, SAA integrates these spatial features by introducing positional encoding and channel-level attention interactions. Finally, SMR is designed to store the semantic information of all samples to enhance the global discriminative ability of the model.

## III. Methodology

The architecture of the proposed ASSMN is shown in Fig. 2. The whole model consists of a MFE, a SAA, a SMR, and a captioning decoder. The ASSMN mainly uses MFE to construct multilevel features at different scales and input them into the SAA, which allows interaction between features at different levels to boost the model to identify key targets at different scales. Subsequently, by using the scale-aware representation as a query, the memory module retrieves the most relevant semantic items in the memory via the attention-based addressing operator, which is finally delivered to the decoder for caption generation. In this section, MFE, SAA, SMR, captioning decoder, and model learning are described in detail.

### A. Multilevel Feature Enhancement

In remote sensing images, objects usually display significant scale variations, where typical attention mechanisms with stacks of static convolutions will result in a limited perception of different objects. Furthermore, with global pooling operations, many existing attention mechanisms fail to preserve critical spatial properties for correlation objects, leading to inaccurate feature enhancement. Compared with the visual-only driven image representations extracted by the ImageNet-pretrained CNN, CLIP-pretrained features driven by visual-verbal alignment pretraining can extract visual features that match the text [9], which alleviates the difficulty of remote sensing image understanding to text generation, resulting in less semantic ambiguity in the generated scene captions.

Inspired by the above ideas, image features are extracted using the visual encoder CLIP-RN50, which has been linguistically aligned with pretrained CLIP. Unlike the previous methods [28], [71], the global attention pooling layer is discarded, and the last three different levels of features are selected as the visual representation of the remote sensing image. This enables the capture of targets at different scales

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in remote sensing images. For an input image $X \in \mathbb{R}^{H\times W\times 3}$, the CLIP-based multilevel extraction process is denoted as

$$F_1, F_2, F_3 = \mathrm{CLIP}_{\mathrm{vision}}(X) \tag{1}$$

where $F_i \in \mathbb{R}^{H_i\times W_i\times D_i}, i \in \{1,2,3\}$ are the features of images with different resolutions $H_i \times W_i$, and $D_i$ is the number of channels for different depth features.

Moreover, to learn richer information from multilevel representation subspaces at different locations, inspired by the multihead self-attention mechanism, we split the separable deformable convolution [31] into four individual groups, where each group of this convolution will have different patterns of spatial aggregation, leading to more robust representations of multiscale features. The separable deformable convolution is formulated as follows:

$$y(p_0) = \sum_{g=1}^{G}\sum_{k=1}^{K} w_g m_{gk} x_g\left(p_0 + p_k + \Delta p_{gk}\right) \tag{2}$$

where $K$ represents the total number of sampling points, and $p_k$ is the $k$th location of the predefined grid sampling points $\{(-1,-1),(-1,0),\ldots,(0,+1),\ldots,(+1,+1)\}$. $G$ denotes the number of convolution groups, and set it to 4. For the $g$th group, $w_g \epsilon R^{C\times C'}$ is the location-unrelated projection weights of the group, and $C' = C/G$ is the group dimension. $m_{gk}$ means the Softmax-normalized modulation scalar of the $k$th point within the $g$th group. $x_g \epsilon R^{C'\times H\times W}$ represents the sliced input feature map, and $\Delta p_{gk}$ is the dynamic offset related to the sampling location $p_k$ in the $g$th group.

To better deal with features of different scales, we combine dilation convolution and deformable convolution and use different stride sizes and dilation rates for feature maps of different depths. While preserving as much spatial detail as possible, feature maps of the same dimensions are obtained for further fusion. The overall process can be expressed as

$$F_i' = \mathrm{DDC}(F_i), \quad i = 1,2,3 \tag{3}$$

where $F_i' \in \mathbb{R}^{H_1\times W_1\times D_1}, i \in \{1,2,3\}$ denote the output feature map. DDC$(\cdot)$ is dilation deformable convolution operation. Then, the equivalent scale features are concatenated along the channel dimension to obtain the complete feature representation $F_4$. With DDC, the object features will be adaptively and dynamically enhanced regardless of scale variations, thus, yielding more robust visual representation.

### B. Scale-Aware Attention Aggregator

The deep features are generated with large receptive fields that capture richer contextual information, and accessing these features enables better prediction using useful contextual information. Similarly, the shallow features have much finer details and better location information, and accessing these features better utilizes native spatial information for generation. Based on these ideas, we propose to bring together all levels of features and fuse them to form a complete semantic representation of the current image. The first stage of SAA involves adding positional encoding to the feature maps to encode the spatial positional information explicitly. The features are then flattened to a sequence along the spatial dimension

$$F_4' = \mathrm{Flatten}(F_4 + \mathrm{PE}(N)) \tag{4}$$

where $F_4' \in \mathbb{R}^{U\times D}$, PE is sin-cosine positional encoding, and $N$ is the spatial position.

Then, we introduce a channelwise attention module to encourage features to focus on channels that benefit most. For that reason, $M$ soft attentions are operated on $F_4'$ to generate multiple critical semantics weights. It is the prerequisite to build $M$ attention maps, which are denoted as $\gamma \in \mathbb{R}^{U\times M}$, $M$ is the space size $H_1 \times W_1$ before $F_4'$ flattening. The weighting coefficient at the location of $u$ in the $m$th attention map is calculated by

$$\gamma_{[u,m]} = \frac{1}{D}\sum_{k=0}^{D} F'_{4_{[u,k]}} * w_{[m,k]} \tag{5}$$

where $\gamma_{[u,m]}$ is the intermediate weighting map, and $w_{[m,k]}$ projects $D$ feature channels into $M$ attention channels. It is then scaled by the SoftMax operation to obtain $\bar{\gamma}_{[u,m]}$ as follows:

$$\bar{\gamma}_{[u,m]} = \frac{\exp\left(\gamma_{[u,m]}\right)}{\sum_{u=0}^{U}\exp\left(\gamma_{[u,m]}\right)}. \tag{6}$$

Finally, the semantic vectors are generated by summing over the position indices $U$ through global averaging. The $k$th element in the $m$th vector of $L$ is calculated by

$$L_{[m,k]} = \frac{1}{U}\sum_{u=0}^{U} F'_{4_{[u,k]}} \otimes \bar{\gamma}_{[u,m]} \tag{7}$$

where $\otimes$ denotes Hadamard product. By aggregating these different level feature representations, we acquire rich semantic information from remote sensing images.

### C. Semantic Memory Reinforcement

Currently, remote sensing image captioning relies on Transformer-based architectures for extracting the semantics in an image and translating it into linguistically coherent descriptions. Although successful, the attention operator only considers a weighted summation of projections of the current input sample, therefore ignoring the relevant semantic information that can come from the joint observation of other samples, that is, rich semantic relations between pixels across different images. In this way, we devise an SMR that can perform attention over activations obtained while processing other training samples. Our SMR models the distribution of past features through the definition of prototype vectors, which are both discriminative and compact.

The semantic memory is designed as a matrix $S \in \mathbb{R}^{O\times Q}$ containing $O$ real-valued vectors of fixed dimension $Q$. For convenience, we assume that $Q$ is the same to the dimension of $L$. Let the row vector $s_i$, $\forall i \in \{1,\ldots,O\}$ denote the $i$th row of $S$. Each $s_i$ denotes a semantic memory item. Given a semantic query $l \in \mathbb{R}^Q$, the semantic memory network obtains $z$ relying on a soft addressing vector $E \in \mathbb{R}^{1\times O}$ as follows:

$$z = ES = \sum_{i=1}^{O} e_i s_i \tag{8}$$

where $E$ is a row vector with nonnegative entries that sum to one. The weight vector $E$ is computed according to $l$. As shown in (8), the addressing weight $E$ is required for

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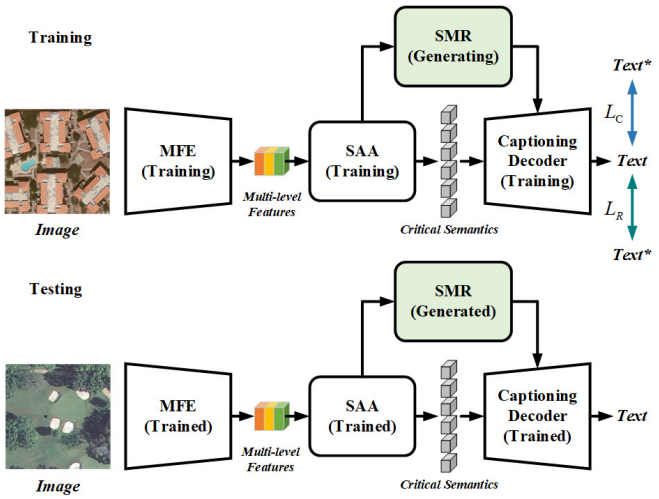



Fig. 3. Training and testing process of ASSMN. The MFE, SAA, and SMR refer to the multilevel feature enhancement module, the SAA, and the SMR module, respectively. The contents in the brackets represent the status of the current modules. Text* denotes ground-truth sentence.

accessing the semantic memory. The hyperparameter $O$ defines the maximum capacity of the semantic memory.

In SMR, the semantic memory $S$ is designed to record explicitly the prototypical sample semantics during training. We define SMR as a content-addressable memory [77] with an addressing scheme that computes attention weights $E$ based on the similarity of the semantic memory items and the query $l$. Each weight $e_i$ is calculated by a SoftMax operation

$$e_i = \frac{\exp\left(d\left(l, e_i\right)\right)}{\sum_{i=1}^{O} \exp\left(d\left(l, e_i\right)\right)} \tag{9}$$

where $d(\cdot, \cdot)$ denotes a cosine similarity operation.

Hence, the SMR retrieves the memory items most similar to $l$ to obtain the representation $z$. To further improve the ability of the model to discriminate between samples, a limited number of relevant semantics are used for captioning in SMR. The attention-based addressing tends to approach this naturally [78]. However, some irrelevant semantics may still be evoked with a complex combination of the memory items via a dense $E$ containing many small elements. To alleviate this issue, a hard shrinkage operation is adopted to promote the sparsity of $E$

$$\hat{e}_i = \frac{\max\left(e_i - \lambda, 0\right) \cdot e_i}{|e_i - \lambda| + \epsilon} \tag{10}$$

where $\hat{e}_i$ denotes the $i$th entry of the semantic memory addressing weight vector $\hat{E}$ after shrinkage. The $\max(\cdot, 0)$ is known as the ReLU function, and $\epsilon$ is a very small positive scalar. In practice setting, the shrinkage threshold $\lambda$ as a value $2/O$. After the shrinkage, we re-normalize $\hat{E}$ by letting $\hat{e}_i = \hat{e}_i / \|\hat{E}\|_1$. The related representation $z$ will be obtained via $z = \hat{E}S$. The sparse addressing strategy encourages the model to employ fewer semantic memory items while simultaneously prioritizing greater relevance thereby generating semantic memory representations with stronger global discriminability.

Due to the restricted memory size and the sparse addressing technique, only a small number of semantic memory items can be addressed every time. As shown in Fig. 3, the state of each module during the training and testing phases of the model is given. Considering the generated semantic memory bank, only the relevant semantics in the memory bank can be retrieved for generation. Thus, the test sample captions can naturally be generated well through the relevant semantic memory.

### D. Captioning Decoder

As can be seen from Fig. 2, the standard transformer [16] is followed, where the decoder is composed of a stack of $V$ identical layers. Each decoder layer is composed of a masked self-attention, a cross-attention, and an FFN layer. We use a residual connection and layer normalization around each sublayer. Given the generated sentence denoted as $T$, get the feature vector $P$ embedded by the word2vec word embedding method and containing positional encoding, which is represented as follows:

$$P = \text{Embeding}\left(T\right) + \text{Postion}\left(N\right). \tag{11}$$

The input to each decoding layer consists of the output of the previous decoding layer and the semantic features generated by SAA and SMR, where the semantic features are injected into the decoder with a cross-attention mechanism. The semantic features $S$ are computed as follows:

$$S = \text{ReLU}\left(\left(Z + L\right) I + B\right) \tag{12}$$

where $I \in \mathbb{R}^{D \times D}$ is the learnable parameter matrix and $B \in \mathbb{R}^{D}$ is the bias. $L$ and $Z$ represent the semantics of the current scene and the supplementary semantics retrieved in the SMR, respectively. Finally, the output $F_d$ of the last decoding layer is regarded as the decoder output. Subsequently, $F_d$ is fed into a fully connected layer and a Softmax function to calculate the probability over a vocabulary of possible words.

### E. Training and Objectives

For the model training, following a standard practice in image captioning [10], the ASSMN is optimized to minimize a word-level cross-entropy loss. The memory contents are simultaneously updated to record the prototypical elements of the encoded global training sets. Given a testing sample, the model performs generation using a restricted number of the correlated semantics recorded in the memory. As a result, the model has a better global discriminative ability. Specifically, a given dataset of paired images and captions $\{x^i, c^i\}_{i=1}^{N}$, our goal is to learn the generation of a meaningful caption for an unseen input image. We can refer to the captions as a sequence of tokens $c^i = c_1^i, \ldots, c_l^i$, where we pad tokens to a maximal length $j$. Training objective is the following:

$$L_c\left(\theta\right) = \max \sum_{i=1}^{N} \log p_\theta\left(c_1^i, \ldots, c_j^i \middle| x^i\right) \tag{13}$$

where $\theta$ denotes the variable parameter of the model. The captioning decoder is trained using parallel computation for enhanced learning efficiency.

After cross-entropy training, self-critical sequence reinforcement learning [42] is further used for image captioning tasks. It incorporates inference into the training process and introduces the evaluation metric CIDEr as a reward function

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TABLE I
GENERAL INFORMATION ABOUT THE DATASETS FOR REMOTE SENSING IMAGE CAPTIONING

| Dataset | Scene categories | Number of images | Number of sentences | Image size |
|---|---|---|---|---|
| Sydney-Captions | 7 | 613 | 3065 | 500×500 |
| UCM-Captions | 21 | 2100 | 10500 | 256×256 |
| NWPU-Captions | 45 | 31500 | 157500 | 256×256 |

in the loss function. This approach helps to cope with the gap between the training and testing phases, while the model learns to generate words based on previous predictions [27], improving its performance in sequential prediction. The reward and loss functions are constructed as follows:

$$\nabla_\theta L_R(\theta) = -\frac{1}{k}\sum_{i=1}^{k}\left(\left(r\left(w^i\right) - b\right)\nabla_\theta \log p_\theta\left(w^i\right)\right) \tag{14}$$

where $k$ is the number of samples, $w^i$ represents the $i$th sentence sampled in the beam, $r(\cdot)$ indicates the reward function, and $b = 1/k\sum_{i=1}^{k} r(w^i)$ is the baseline reward, which is introduced to insure the stability of training.

## IV. EXPERIMENTAL STUDY

### A. Experimental Settings and Datasets

To verify the performance of the proposed ASSMN, we conduct experiments on three public datasets in Table I, including the Sydney-Captions, UCM-Captions, and NWPU-Captions. Every image is annotated with five reference sentences for three datasets.

*Sydney-Captions [65]:* This is grounded in the Sydney dataset, which belongs to remote sensing scene datasets. It encompasses 613 images that are classified into seven distinct scenes: rivers, residential areas, airports, grasslands, oceans, industrial zones, and runways. Each image has dimensions of 500 × 500 pixels, and the pixel resolution is approximately 50 cm. All these images are painstakingly extracted from Sydney-related images that are obtained by downloading from Google Earth.

*UCM-Captions [65]:* It is grounded in the UC Merced remote sensing scene classification dataset of land use data, which cover 21 typical land-use scene categories, including agriculture, airplanes, baseball stadiums, beaches, buildings, brush, dense residential, forests, highways, golf courses, rivers, harbors, intersections, medium-density residential, overpasses, mobile home parks, parking lots, runways, sparse residences, oil storage tanks, and tennis courts. Each category has 100 images with an RGB spatial pixel resolution of 30 cm. The UCM dataset is extracted from aerial orthophotos and downloaded from the USGS.

*NWPU-Captions [20]:* It builds on the remote sensing image classification dataset created by Northwestern Polytechnical University, which contains 45 classes: airplane, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, residential, mobile home park, mountain, overpass, palace, parking lot, railway, station, rectangular, etc. The ground sampling distance of each image ranges from 0.2 to 30 m. The NWPU dataset is richer in classes and quantities.

The ASSMN uses the CLIP pretrained [29] and frozen ResNet-50 network as an image feature extractor. The specific three scales of features chosen are from the output of the conv3_x, conv4_x, and conv5_x layers. For the three datasets, the number of memory bank O is chosen as 500, 1000, and 2000, respectively, with an embedding dimension of 512. The captioning decoder follows the vanilla transformer architecture with three cycles and head = 8 parallel attention layers in the multihead attention mechanism. For text generation, a start token ⟨bos⟩ and an end token ⟨eos⟩ are added before and after each ground truth sequence. The word embedding dimension is configured as 512. The maximum output sequence length is 25. We set the batch size to 50 and optimized the model parameters using the Adam optimizer. The hidden dimension of each FFN layer is set to 2048. In addition, we defined "patience" to monitor the model's performance on the validation set and to decide whether the model needed to be improved.

During the cross-entropy training process, we adopt the learning rate scheduling strategy [71], which incorporates a warm-up operation comprising 10 000 iterations. We keep the model with the highest CIDEr score on the validation set as an initialization model for subsequent training phases or inference. When the patience value reaches 5, the training focus shifts to CIDEr optimization, and the learning rate is fixed at $5 \times 10^{-6}$. In both the optimization and decoding phases, we use a beam size of five to generate candidate sequences. All experiments are conducted on NVIDIA GeForce RTX 2080Ti-11 GB.

### B. Evaluation Criteria

In the past, image captioning evaluation heavily relied on machine translation evaluation metrics such as BLEU [79], recall-oriented understudy for gisting evaluation (ROUGE) [80], and METEOR [81], which lack relevance to human assessment. More recently, with the introduction of CIDEr [82] and semantic propositional image caption evaluation (SPICE) [83] indicators designed explicitly for image captioning, this issue has been addressed. To assess the model's performance from various perspectives, the metrics BLEU, ROUGE, METEOR, CIDEr, and SPICE are selected for captioning assessment. We provide a detailed introduction to these five metrics below and organize them in Table II.

1) *BLEU:* Bilingual evaluation understudy (BLUE) is a commonly used automatic evaluation metric to measure the similarity between generated text and reference text. It focuses more on the precision rate and calculates the score by calculating the proportion of n-grams (from single to multiple words) that appear in the reference text in the generated text, and then combines exact and partial matches. Here, n-gram is a sequence of n continuous words with values ranging from 1 to 4.
2) *ROUGE:* ROUGE is an evaluation metric to measure the recall between generated text and reference text. It

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evaluates the similarity between the generated text and the reference text by comparing the length of the longest common subsequence.

3) *METEOR:* Metric for evaluation of translation with explicit ordering (METEOR) combines exact and inexact matching criteria such as synonyms and stems. It uses external dictionaries to help identify synonyms and stems, allowing for a more overall assessment of the quality of the generated text.
4) *CIDEr:* Consensus-based image description evaluation (CIDEr) focuses on the diversity and consistency of generated captions. It calculates scores by comparing word frequency statistics between the generated text and multiple reference texts, emphasizing consistency across reference texts.
5) *SPICE:* SPICE is highly correlated with human judgment. Unlike traditional n-gram-based evaluation methods that mainly evaluate by calculating the lexical overlap between the reference text and the generated text, SPICE uses semantic parse trees to measure the similarity between generated and reference texts, providing better semantic matching.

### C. Compared Methods

To evaluate the effectiveness of the ASSMN, we compare our method with several state-of-the-art methods as below.

Soft-attention and hard-attention [73] introduce hard attention and soft attention mechanisms into the CNN-RNN framework, respectively. They employ the pretrained VGG-16 model on ImageNet datasets as the visual extraction backbone network.

FC-Att+LSTM and SM-Att+LSTM [67] integrate low-level features and high-level attribute features based on the attribute attention mechanism. They use the VGG-16 model, which is not affirmed to be pretrained on a specific dataset, as the backbone network for visual extraction.

Trans+SCST [74] proposes a new transformer structure. It added an adaptive feature fusion on the transformer and reinforcement learning is used to improve the actual performance of the model. It employs the pretrained ResNet-101 model on ImageNet datasets as the visual extraction backbone network.

MC-Net [22] contains a CNN image encoder enhanced with a multiscale extraction module, a feature fusion module, and a fine-tuned LSTM equipped with a visual-text alignment module. It uses the VGG-16 model, which is not affirmed to be pretrained on a specific dataset, as the backbone network for visual extraction.

PKG-transformer [7] first enriches the object and scene features by leveraging the object–scene relationship. It then integrates the scene–object relationship into the transformer encoder as prior knowledge in a self-attention manner. It employs a pretrained ResNet-152 model on the ImageNet dataset, and uses a pretrained Faster R-CNN with manually labeled target detection boxes as the backbone network for visual extraction.

FPHD [75] designs a deformable transformer by separating foreground and background based on the deeplabV3+ model and learns multiscale features from foreground and background through attention interaction. It employs a pretrained ResNet-101 model on the ImageNet dataset and uses a pretrained DeepLabV3+ on the iSAID dataset as the backbone network for feature extraction.

RS-CapRet [76] uses CLIP encoding as a prefix for captions and generates image captions and text-image retrieval by using a simple mapping network and then fine-tuning the LLamaV2-7B model. It employs a pretrained CLIP image encoder with ViT-L/14 as the backbone network for feature extraction.

MLAT [23] utilizes LSTM to fuse features from different layers in ResNet-50 to extract multiscale information. The captions are then generated by a multilayer aggregation transformer. It uses the ResNet-50 model that was not pretrained on any specific dataset as the backbone network for visual extraction.

HCNet [19] uses CNN for hierarchical visual feature extraction and alignment of visual and textual features by contrast loss and a dual-LSTM decoder for language generation. It employs the pretrained ResNet-50 model on ImageNet datasets as the visual extraction backbone network.

MG-transformer [71] integrates CLIP and CNN features for a comprehensive image representation and then utilizes a global grouping attention mechanism to establish connections between visual and textual features at different scales. It employs a pretrained ResNet-152 model on the ImageNet dataset and uses a pretrained CLIP image encoder with ViT-B/32 as the backbone network for feature extraction.

VRoPE-transformer [26] extracts features with a pretrained object detector and ResNet, and it designs a visual rotational position encoding module to model the orientation relationships between objects. It employs a pretrained ResNet-152 model on the ImageNet dataset and uses a fine-tuned Redet with manually labeled target detection boxes as the backbone network for visual extraction.

ICNET [25] extracts local features through designing concept mapping network and enhances relational understanding via a global feature enhancement module. It adopts ResNet-152 pretrained on ImageNet to extract region features and the pretrained CLIP image encoder with ViT-b/32 backbone to extract global features. SFDR [84] integrates the semantic–spatial feature fusion and dynamic graph feature refinement. The semantic–spatial feature fusion utilizes a multilevel feature representation strategy by leveraging pretrained CLIP features on ViT-B/32, grid features on ResNet-152, and ROI features on faster R-CNN to integrate rich semantic and spatial information.

### D. Comparison With Existing Methods

This section provides a comparison of the proposed ASSMN with several state-of-the-art methods, including single visual encoder-based methods and hybrid visual encoder-based methods. The experimental results on three datasets are presented in Tables III–V.

*1) Result on Sydney-Captions:* As shown in Table III, methods based on hybrid visual encoders have achieved significant performance improvements by employing different types of visual feature extractors. For example, PKG-transformer and VRoPE-transformer outperform single-visual-encoder methods like HCNet. However, methods using hybrid visual encoders require additional object-related manual labels for

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TABLE II

EVALUATION METRICS FOR REMOTE SENSING IMAGE CAPTIONING. THE TABLE PRESENTS A COMPREHENSIVE OVERVIEW OF VARIOUS EVALUATION METRICS EMPLOYED FOR ASSESSING THE PERFORMANCE OF REMOTE SENSING IMAGE CAPTIONING MODELS

| Metric | Based On | Original Task | Advantages | Drawbacks |
|---|---|---|---|---|
| BLEU | Precision | Machine translation | Simple and efficient | Grammar issues and limited by text length |
| ROUGE | Recall | Machine text summarization | Simple and orderly | Disregard synonyms |
| METEOR | Precision Recall Pen | Machine translation | Consider stems and synonyms | Reliance on external knowledge sources |
| CIDEr | TF-IDF | Image caption | Consider global semantic information | Hard to optimize |
| SPICE | F-score | Image caption | Effective for evaluating complex scenario descriptions | Dependency semantic analysis tools |

TABLE III

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THE ARTICLE

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Soft-attention [73] | 73.22 | 66.74 | 62.23 | 58.20 | 39.42 | 71.27 | 249.93 | - |
| Hard-attention [73] | 75.91 | 66.10 | 58.89 | 52.58 | 38.98 | 71.89 | 218.19 | - |
| FC-Att+LSTM [67] | 80.76 | 71.60 | 62.76 | 55.44 | 40.99 | 71.14 | 220.33 | - |
| SM-Att+LSTM [67] | 81.43 | 73.51 | 65.86 | 58.06 | 41.11 | 71.95 | 230.21 | - |
| Trans+SCST [74] | 80.16 | 70.95 | 63.14 | 57.24 | 41.83 | 72.62 | 253.92 | - |
| MC-Net [22] | 83.40 | 75.00 | 67.80 | 60.70 | 40.60 | 73.90 | 256.40 | 45.60 |
| PKG-Transformer [7] | 83.17 | 77.83 | 72.84 | 68.24 | 45.28 | 77.06 | 284.76 | 44.05 |
| FPHD [75] | 83.73 | 77.71 | 71.98 | 66.59 | 45.48 | 78.60 | 303.69 | **48.39** |
| RS-CapRet [76] | 78.70 | 70.00 | 62.80 | 56.40 | 38.80 | 70.70 | 239.20 | 42.30 |
| MLAT [23] | 86.29 | 75.85 | 67.48 | 60.33 | 41.51 | 76.11 | 255.75 | 39.04 |
| HCNet [19] | 76.86 | 71.09 | 65.73 | 61.02 | 39.80 | 71.72 | 247.14 | - |
| MG-Transformer [71] | 86.68 | 80.87 | 75.88 | 71.22 | 47.70 | 79.70 | 327.76 | 48.09 |
| VRoPE-Transformer [26] | 87.30 | 82.37 | 78.31 | 74.38 | **49.79** | **81.54** | **343.88** | 48.22 |
| ICNET [25] | 84.68 | 79.85 | 75.61 | 71.50 | 47.40 | 77.91 | 302.58 | 47.28 |
| SFDR [84] | 85.20 | 80.18 | 75.69 | 71.72 | 47.34 | 79.74 | 328.64 | 45.22 |
| ASSMN | **88.00** | **83.52** | **79.51** | **75.18** | 46.53 | 80.00 | 315.69 | 45.04 |

TABLE IV

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THE ARTICLE

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Soft-attention [73] | 74.54 | 65.45 | 58.55 | 52.50 | 38.86 | 72.37 | 261.24 | - |
| Hard-attention [73] | 81.57 | 73.12 | 67.02 | 61.82 | 42.63 | 76.98 | 299.47 | - |
| FC-Att+LSTM [67] | 81.35 | 75.02 | 68.49 | 63.52 | 41.73 | 75.04 | 299.58 | - |
| SM-Att+LSTM [67] | 81.54 | 75.75 | 69.36 | 64.58 | 42.40 | 76.32 | 318.64 | - |
| Trans+SCST [74] | 83.82 | 79.01 | 74.43 | 70.11 | 44.64 | 77.63 | 356.53 | - |
| MC-Net [22] | 84.50 | 78.40 | 73.20 | 67.90 | 44.90 | 78.60 | 335.50 | 47.30 |
| PKG-Transformer [7] | 90.48 | 87.04 | 84.10 | 81.39 | 54.66 | 86.57 | 427.49 | 57.01 |
| FPHD [75] | 82.30 | 77.00 | 72.28 | 67.92 | 44.39 | 78.39 | 346.29 | 48.25 |
| RS-CapRet [76] | 84.30 | 77.90 | 72.20 | 67.00 | 47.20 | 81.70 | 354.80 | 52.50 |
| MLAT [23] | 89.96 | 83.19 | 77.52 | 72.02 | 48.27 | 84.09 | 305.01 | 25.93 |
| HCNet [19] | 88.26 | 83.35 | 78.85 | 74.49 | 48.65 | 83.91 | 351.83 | - |
| MG-Transformer [71] | 91.28 | 88.11 | 85.37 | 82.95 | 56.01 | 88.77 | 448.39 | **58.39** |
| VRoPE-Transformer [26] | 89.81 | 86.45 | 83.56 | 80.98 | 53.36 | 85.93 | 434.63 | 55.38 |
| ICNET [25] | 91.02 | 87.52 | 84.31 | 81.35 | 55.25 | 87.85 | 435.66 | 58.18 |
| SFDR [84] | 91.32 | 87.96 | 85.10 | 82.31 | 54.76 | 88.40 | 448.76 | 57.58 |
| ASSMN | **91.70** | **88.78** | **86.22** | **83.97** | **56.37** | **89.08** | **451.26** | 58.09 |

the extra visual extractors, which increase data preprocessing costs. The proposed method addresses this issue by designing a memory module on top of a single visual encoder, effectively integrating both current and historical sample information. Consequently, it significantly narrows the performance gap compared to methods based on hybrid visual encoders.

*2) Result on UCM-Captions:* As shown in Table IV, the proposed method demonstrates a more obvious performance advantage over the two existing remote sensing image captioning methods on the UCM-captions dataset. For example, on the ROUGE metric, the proposed method achieves a 0.68% improvement over SFDR and a 5.17% improvement over HCNet. Compared to single visual encoder-based

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TABLE V

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THE ARTICLE

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Soft-attention [73] | 73.10 | 60.90 | 52.50 | 46.20 | 33.90 | 59.90 | 113.60 | - |
| Hard-attention [73] | 73.30 | 61.00 | 52.70 | 46.40 | 34.00 | 60.00 | 110.30 | - |
| FC-Att+LSTM [67] | 73.60 | 61.50 | 53.20 | 46.90 | 33.80 | 60.00 | 123.10 | - |
| SM-Att+LSTM [67] | 73.90 | 61.70 | 53.20 | 46.80 | 33.00 | 59.30 | 123.60 | - |
| Trans+SCST [74] | 86.96 | 78.66 | 72.39 | 67.34 | 43.58 | 76.60 | 190.68 | - |
| MC-Net [22] | 74.10 | 62.60 | 54.40 | 47.80 | 34.70 | 61.10 | 115.90 | 28.90 |
| PKG-Transformer [7] | 88.30 | 80.40 | 74.00 | 68.90 | 44.40 | 77.30 | 197.70 | 28.90 |
| FPHD [75] | 75.15 | 62.91 | 54.57 | 48.28 | 31.87 | 58.58 | 120.71 | 26.78 |
| RS-CapRet [76] | 87.10 | 78.70 | 71.70 | 65.60 | 43.60 | 77.60 | 192.90 | **31.10** |
| MLAT [23] | 89.22 | 77.72 | 68.76 | 61.43 | 42.54 | 77.53 | 155.45 | 14.36 |
| HCNet [19] | 89.54 | 82.51 | 76.58 | 71.68 | **47.42** | **80.53** | 209.27 | - |
| MG-Transformer [71] | 90.30 | 83.04 | 76.90 | 71.63 | 42.91 | 77.26 | 213.48 | 27.19 |
| VRoPE-Transformer [26] | 90.50 | 83.50 | 77.60 | 72.70 | 44.60 | 78.60 | 210.80 | 28.50 |
| ICNET [25] | 91.39 | 84.18 | 77.97 | 72.79 | 45.02 | 79.47 | 215.02 | 28.43 |
| SFDR [84] | 89.03 | 80.39 | 73.00 | 66.94 | 41.59 | 75.21 | 205.51 | 26.60 |
| ASSMN | **91.67** | **84.55** | **78.45** | **73.33** | 44.19 | 78.72 | **216.47** | 28.29 |

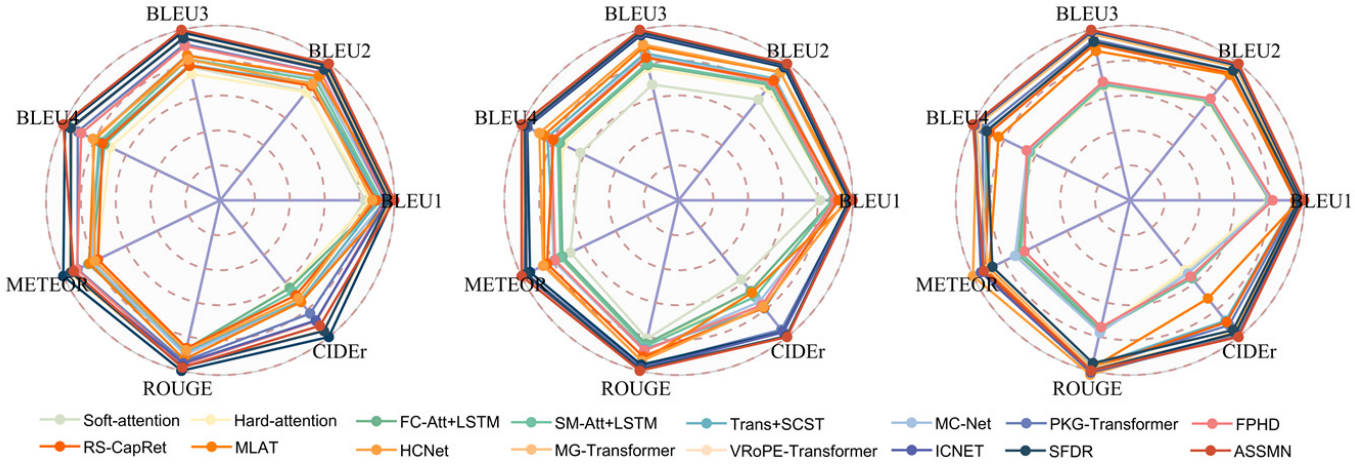



Fig. 4. Comparison results of state-of-the-art method on three datasets.

methods, hybrid visual encoder-based methods achieve high performance. It can be observed that methods based on multiscale CNN features achieve higher performance compared to those relying solely on traditional single-layer CNN features. For instance, on the CIDEr metric, MC-Net outperforms SM-Att+LSTM by 16.86%. Furthermore, when compared to methods based purely on the Transformer framework, multiscale CNN-based approaches also demonstrate solid performance.

*3) Result on NWPU-Captions:* Table V demonstrates that for the large-scale, diverse-resolution, and multicategory NWPU-captions dataset, the ASSMN enhances the understanding of the model on the current scene due to storing the semantic features of the historical samples based on the single visual encoder-based methods. In most of the metrics, ASSMN obtained the best experimental metrics beyond the hybrid visual encoder-based methods. This is because our model not only fully utilizes the features of multilevel image text alignment to retain historical sample information that is useful for subsequent text generation but also improves it by utilizing an SAA. This improves the ability of the model to perceive current and historical global features of objects. In particular, HCNet outperforms methods based on hybrid visual encoders to achieve optimal results on both METEOR and ROUGE metrics. This success is largely attributable to its introduction of a visual-linguistic contrastive loss. Remarkably, our approach surpasses most hybrid visual encoder-based methods even without employing contrastive losses. In addition, Fig. 4 graphically gives a contrasting radar chart of the different algorithms. This proves the superiority of ASSMN.

## E. Ablation Study

The contributions of each module to the overall performance of ASSMN are evaluated through an ablation study. Specifically, the proposed ASSMN with only the CLIP-RN50 and captioning decoder is used as the baseline model. The three modules are added one by one to construct ablation models. Additionally, an ablation study is conducted on the number of memory bank. Furthermore, through complexity and efficiency experiments, the parameters, FLOPs, and inference speed of ASSMN and its ablation models are analyzed and compared with several state-of-the-art methods. Finally, several common pretrained visual extractors are compared. Specific experimental results are provided below.

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TABLE VI
RESULT OF ABLATION STUDY ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Base | 85.33 | 79.39 | 73.99 | 69.36 | 47.09 | 77.91 | 285.57 | 42.76 |
| Base+MFE | 84.41 | 78.85 | 73.80 | 69.35 | **47.74** | 77.84 | 309.13 | 44.91 |
| Base+MFE+SAA | 85.83 | 81.12 | 76.68 | 72.82 | 47.44 | 79.21 | 290.94 | 44.52 |
| ASSMN (Base+MFE+SAA+SMR) | **88.00** | **83.52** | **79.51** | **75.18** | 46.53 | **80.00** | **315.69** | **45.04** |

TABLE VII
RESULT OF ABLATION STUDY ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Base | 88.63 | 84.81 | 81.78 | 78.91 | 53.45 | 86.08 | 412.42 | 54.82 |
| Base+MFE | 88.87 | 85.17 | 82.07 | 79.43 | 52.31 | 85.05 | 419.44 | 55.81 |
| Base+MFE+SAA | 88.91 | 84.98 | 81.65 | 78.47 | 52.06 | 84.93 | 397.26 | 55.06 |
| ASSMN (Base+MFE+SAA+SMR) | **91.70** | **88.78** | **86.22** | **83.97** | **56.37** | **89.08** | **451.26** | **58.09** |

TABLE VIII
RESULT OF ABLATION STUDY ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Method | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| Base | 86.98 | 76.60 | 68.43 | 61.96 | 37.69 | 70.69 | 190.79 | 24.06 |
| Base+MFE | 87.68 | 77.83 | 70.05 | 63.69 | 37.90 | 70.86 | 193.30 | 23.32 |
| Base+MFE+SAA | 91.01 | 83.33 | 76.55 | 70.91 | 41.94 | 76.80 | 211.18 | 26.26 |
| ASSMN (Base+MFE+SAA+SMR) | **91.67** | **84.55** | **78.45** | **73.33** | **44.19** | **78.72** | **216.47** | **28.29** |

1) *MFE:* The role of MFE in the model is explored through the Base and Base+MFE ablation models in Tables VI–VIII. The MFE module is used to augment the CLIP-extracted multilevel features by a separable deformation convolution for further extraction of key features at different scales. For the base model, we directly apply the last level of features extracted by CLIP for the language decoder to generate language descriptions. The results of comparing the two sets of data show that Base+MFE outperforms the base model. Due to the introduction of MFE, the model can have a more complete representation of features at different scales. This also verifies that utilizing separable deformation convolution improves the performance of the model while ensuring efficiency.
2) *SAA:* Likewise, the impact of SAA on different datasets is investigated in Tables VI–VIII. A comparison of the Base+MFE and Base+MFE+SAA models shows that with the implementation of the SAA module, effective aggregation of multilevel features is achieved, resulting in improved model performance. Among other things, SAA selectively emphasizes important information in different levels of features, providing a more robust global representation with more discriminative properties for the present sample.
3) *SMR:* The effectiveness of SMR is validated on three datasets. ASSMN is based on the baseline and contains all the components mentioned above. Compared to the Base+MFE+SAA model, which does not contain the SMR, the fact that ASSMN stores information about all the samples during the training process makes it possible to fully utilize the information from historically relevant samples during the generation process, as opposed to extracting relevant information for generating descriptions from the current samples only. This strategy enhances the interaction between the semantic features of the current sample and the features of all samples, resulting in a well-tuned representation ability. Tables VI–VIII give a comparison of on different datasets. It is worth noting that on the Sydney-captions, ASSMN's performance on the METEOR metric falls below that of Base+MFE. This may be attributed to the relatively small scale of the Sydney-captions, where the SMR module stores less-relevant information. However, on larger datasets such as UCM-captions and NWPU-captions, the advantages of SMR become more pronounced.
4) *Number of Memory Bank:* In this section, we perform a study on the number of memory banks chosen for the SMR on three datasets. Tables IX–XI list the experimental results, and it can be seen that there is a significant difference in the dimensions of the SMR for datasets of different scales. For the Sydney-captions dataset, choosing a memory dimension of 500 is a better choice, while for the UCM-captions dataset and the NWPU-captions dataset, choosing dimensions of 1000 and 2000, respectively. It is not difficult to find out that this is also directly proportional to the size of the three datasets themselves. For example, the Sydney-captions dataset contains only seven scene classes, indicating that it contains relatively few targets and correlations, and the selection of a memory number of 500 is

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TABLE IX
RESULT OF MEMORY SIZE ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Number of memory bank | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| 500 | **88.00** | **83.52** | **79.51** | **75.18** | **46.53** | **80.00** | **315.69** | 45.04 |
| 1000 | 86.81 | 80.32 | 75.00 | 70.18 | 45.33 | 77.92 | 280.48 | 43.55 |
| 1500 | 86.08 | 80.06 | 74.75 | 69.91 | 45.05 | 77.47 | 294.03 | 44.21 |
| 2000 | 84.38 | 78.82 | 73.94 | 69.86 | 46.44 | 78.68 | 303.19 | **45.67** |
| 2500 | 85.97 | 80.16 | 74.74 | 69.55 | 45.93 | 77.03 | 281.09 | 44.62 |

TABLE X
RESULT OF MEMORY SIZE ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Number of memory bank | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| 500 | 90.01 | 85.82 | 82.26 | 79.04 | 53.89 | 87.83 | 438.40 | 56.93 |
| 1000 | **91.70** | **88.78** | **86.22** | **83.97** | **56.37** | **89.08** | **451.26** | **58.09** |
| 1500 | 89.29 | 86.10 | 83.37 | 80.88 | 51.45 | 84.50 | 418.51 | 53.91 |
| 2000 | 88.76 | 84.79 | 81.31 | 78.06 | 51.61 | 85.15 | 410.33 | 54.99 |
| 2500 | 90.91 | 88.13 | 85.72 | 83.58 | 55.98 | 87.80 | 439.65 | 57.70 |

TABLE XI
RESULT OF MEMORY SIZE ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

| Number of memory bank | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr | SPICE |
|---|---|---|---|---|---|---|---|---|
| 500 | 90.67 | 83.28 | 76.71 | 71.17 | 43.36 | 77.87 | 212.95 | 27.54 |
| 1000 | 91.65 | 84.43 | 78.02 | 72.57 | 44.31 | 79.33 | **217.39** | 27.77 |
| 1500 | 91.16 | 83.89 | 77.65 | 72.42 | **44.75** | 78.81 | 212.30 | 28.32 |
| 2000 | **91.67** | **84.55** | **78.45** | **73.33** | 44.19 | 78.72 | 216.47 | 28.29 |
| 2500 | 91.23 | 84.11 | 77.95 | 72.72 | 45.00 | **79.54** | 214.72 | **29.02** |

TABLE XII
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS IN MODEL PARAMETERS, FLOPS, AND INFERENCE SPEED

| Method | Parameters | FLOPs | Inference Speed |
|---|---|---|---|
| MLAT [28] | 149.77M | - | - |
| PKG-Transformer [23] | 31.94M | 1.58G | 24.56 |
| VRoPETransformer[26] | 31.27M | 1.38G | - |
| MG-Transformer [71] | 38.56M | 1.60G | 31.09 |
| ICNET [25] | 24.18M | 1.04G | 26.63 |
| SFDR [84] | 36.55M | 2.05G | 12.92 |
| Base | 13.77M | 0.41G | 42.37 |
| Base+MFE | 18.50M | 0.64G | 36.59 |
| Base+MFE+SAA | 18.53M | 0.64G | 35.53 |
| ASSMN | 18.79M | 0.65G | 33.17 |

sufficient for storing its target and attribute information. Too large a memory number, on the contrary, leads to learning difficulties and failure to learn the associated semantic information, resulting in a decrease in model performance. For the UCM-Captions dataset, the number of memory bank 1000 is its best choice. Finally, the NWPU-captions dataset shows that when the size of the dataset is enlarged, except for the very small memory scale, all of them can achieve good performance, which reflects that the SMR can also maintain good performance as the data scale increases.

5) *Complexity and Efficiency Analysis:* This section provides a comparison of the computational complexity of the proposed method with state-of-the-art methods, especially the number of parameters, FLOPs, and inference speed (images/second). Table XII shows the comparison of the specific number of parameters, the FLOPs, and inference speed. Additionally, a complexity and efficiency analysis are conducted within the proposed ASSMN framework for models, such as Base, Base+MFE, and Base+MFE+SAA. Compared with the state-of-the-art methods, the ASSMN achieves good results with the minimum number of parameters. Compared to the baseline, ASSMN achieved a more significant improvement in performance with only a slight increase in the number of parameters. It is evident that the increase in the number of parameters proposed by the method primarily stems from the MFE module, which precisely demonstrates that ASSMN possesses more robust feature representation ability. In terms of FLOPs, the proposed method reduces FLOPs by 0.39G compared to ICNET, which has the lowest FLOPs among the comparison methods. Since the proposed method is based on a single encoder, it is evident that ASSMN achieves excellent efficiency. Regarding inference speed, it maintains comparable inference speeds to the comparison methods while still delivering impressive performance.

6) *Comparison of Different Pretrained Visual Backbones:* To demonstrate the effectiveness and superiority of CLIP pretrained features, we experimented with different visual features and analyze ablation results.

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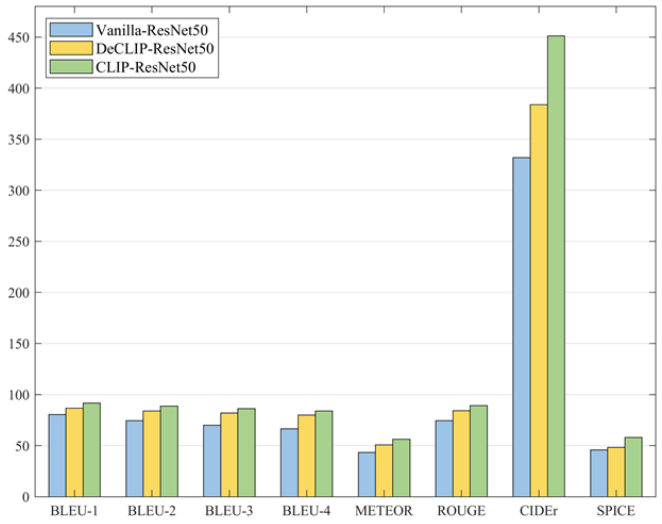



Fig. 5. Comparison results of ASSMN with different pretrained visual backbones on UCM-captions.

Fig. 5 displays ablation experiments on different visual features. For a fair comparison, the experiments use identical settings to compare multiscale visual features extracted from vanilla ResNet-50 pretrained on ImageNet [15], ResNet-50 pretrained on DeCLIP [85], and ResNet-50 pretrained on CLIP [29] across UCM-Captions dataset. Experiments reveal that when visual backbone architectures are identical, different pretraining styles significantly impact performance on downstream tasks. For remote sensing image captioning, adopting a visual language pretraining approach yields superior performance. Compared to visual features using DeCLIP, visual features using CLIP yield better metric results. This is inseparable from CLIP's simple training approach and its large-scale image-text data pairs. This demonstrates that CLIP visual features significantly outperform other visual features, making them beneficial for remote sensing image captioning.

7) *Discussion of Remote Sensing Spatiotemporal Vision-Language Models:* With the upgrading of sensing platforms and increased revisit frequencies, multitemporal remote sensing images have become readily accessible, opening new paths for remote sensing spatiotemporal image interpretation. Remote sensing spatiotemporal vision-language models [86] must comprehend spatial patterns such as object categories and spatial relationships while capturing temporal evolution processes such as emergence, disappearance, and growth. These models integrate natural language modalities with temporal image sequences, enriching remote sensing spatiotemporal image interpretation. ASSMN integrates remote sensing scene information from different times and locations, aligning with spatiotemporal image interpretation objectives. For instance, in remote sensing image change captioning tasks, MFE and SAA can be employed to extract semantics about targets and their attributes from multitemporal remote sensing images. Subsequently, SMR enables the effective joint storage of these sequence elements, thereby allowing for precise change captioning.

### F. Qualitative Analysis

*1) Caption Generation Results:* As shown in Fig. 6, we extracted several caption results generated by our model from three datasets: Sydney-captions, UCM-captions, and NWPU-captions. For comparison, we show three kinds of caption results: results from our model based on memory reinforcement, results from the baseline based on single visual encoder-based methods, and the results of the MG-transformer, which represents the current state-of-the-art based on hybrid visual encoder-based approaches. As can be seen in Fig. 6, the captions generated by ASSMN describe the objects more accurately and with a lower error rate than the baseline approach. Specifically, for the first image presented in Fig. 6, the foreground (road) is hidden in the background (residential area). In this case, generating sentences containing roads with the benchmark method is very challenging and lacks interpretability. The captions generated by the benchmark method do not contain red line information. However, our model can efficiently extract this hidden foreground information and reflect it in the generated sentences. In addition, the appropriateness of the sentences meets the requirements. All these sentence improvements are attributed to the fact that our multilevel feature aggregation and global sample memory can query historically relevant features during semantic feature learning, thus reducing the interference of external noise. The phenomenon is similar to the other captions in Fig. 6. Specifically, the third sample "plants" surrounds four tennis courts, and the number of tennis courts can be accurately identified because the SMR stores semantic information about tennis courts. In contrast, the sentence generated by the benchmark method only uses "many" to obscure the number of tennis courts, while the MG-transformer describes three tennis courts. Similarly, comparing the 9th, 10th, and 11th samples, the ASSMN is found to have the ability to accurately describe the number of counts. This is especially true in the 11th sample since the vessel is close to the target about the "shore" and the target itself is narrow and close in color to the body of water. The proposed MFE incorporates the concept of multiscale and can recognize the number of boats well. This also proves the strong generalization ability of our proposed method. Finally, the superiority of the model is also demonstrated through other samples.

*2) Real-World Applicability:* To further evaluate the performance of the proposed method real-world scenarios, we conducted generalization tests on the model using Sentinel-2 images from the SEN1-2 dataset and Google Earth images from the LoveDA dataset. For inference, we only use the optical images from the Sentinel-2 satellite and Google Earth. Fig. 7 shows the results of our method on Sentinel-2 and Google images. In particular, we tested the model trained on the NWPU-captions dataset using Sentinel-2 and Google Earth images. The choice to train the model on the NWPU-captions stems from the significant resolution disparity between Sentinel-2 and Google Earth images. The NWPU-captions contains images at multiple resolutions, allowing for a degree of evaluation of the trained model's generalization

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GT: Many houses arranged neatly while a highway passed by.
Baseline: Many houses arranged neatly with some grasslands beside.
MG-Transformer: A residential area with many houses arranged neatly and a piece of lawn beside.
ASSMN: A residential area with houses arranged neatly and some roads go through this area.



GT: There are lots of plants on both sides of the river.
Baseline: A small river flows through the dense plants.
MG-Transformer: Some green bushes and small bunkers on the meadow.
ASSMN: Some green bushes and white bunkers on the meadow with some roads.



GT: Some tennis courts on the green ground arranged neatly with some plants surrounded.
Baseline: There are many tennis courts arranged neatly and surrounded by some plants.
MG-Transformer: There are three tennis courts arranged neatly and surrounded by some trees.
ASSMN: There are four tennis courts arranged neatly and surrounded by some plants.



GT: An overpass with a road go across another roads diagonally with some plants surounded.
Baseline: A road go across another roads diagonally with some cars on the roads.
MG-Transformer: A road go across another one diagonally with some cars on the roads.
ASSMN: An overpass with a road go across another roads diagonally with some cars on the roads.



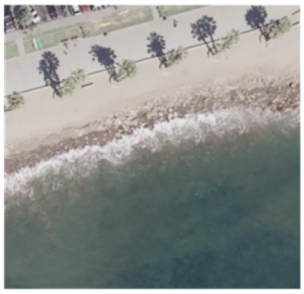
GT: There is an airplane stopped at the airport.
Baseline: An airplane is stopped at the airport.
MG-Transformer: An airplane is surrounded with some cars in the airport.
ASSMN: An airplane with blue fuselage is stopped at the airport.




GT: It is a straight runway with some mark lines and a number on it .
Baseline: A straight runway with some white mark lines on it.
MG-Transformer: A straight runway with some white mark lines and a number and a letter l on it.
ASSMN: A straight runway with some mark lines and a number on it.



GT: Many mobile homes are arranged haphazardly in the mobile home park.
Baseline: A medium residential area with a road goes through this area.
MG-Transformer: Many mobile homes arranged in lines in the mobile home park and some roads go through this area.
ASSMN: Many mobile homes arranged haphazardly in the mobile home park and some roads go through this area.



GT: This is a beach with clear waters and white sands.
Baseline: Waves slapping a white sand beach throw up white foams.
MG-Transformer: Waves are slapping a white sand beach and some plants on the beach.
ASSMN: Waves slapping a white sand beach throw up white foams.



GT: Several buildings are beside the storage tanks.
Baseline: There are four black storage tanks on the bare land.
MG-Transformer: There are several neatly arranged large oil storage tanks in the factory.
ASSMN: There are four storage tanks on the bare land.



GT: The baseball field is surrounded by neat rows of small houses.
Baseline: Two baseball diamonds are surrounded by grass.
MG-Transformer: A baseball diamond next to a parking lot and some buildings beside.
ASSMN: There are two baseball diamonds in the middle of grass next to many buildings.



GT: The two orange ships are docked at the oil terminal.
Baseline: There are many trees beside the ships.
MG-Transformer: There are two ships in the green water on the shore.
ASSMN: There are two ships in the black water on the shore.



GT: A basketball court on the vacant lot and some trees beside.
Baseline: A basketball court on the lawn and some trees beside.
MG-Transformer: Many trees are near the basketball court.
ASSMN: There is a basketball court on the open place next to some trees.

Fig. 6. Example sentences generated by Baseline, MG-transformer, and ASSMN methods on three datasets. The GT denotes the ground truth sentence. The words in red incorrectly matched the corresponding image, while some crucial objects and relationship words are highlighted in green.

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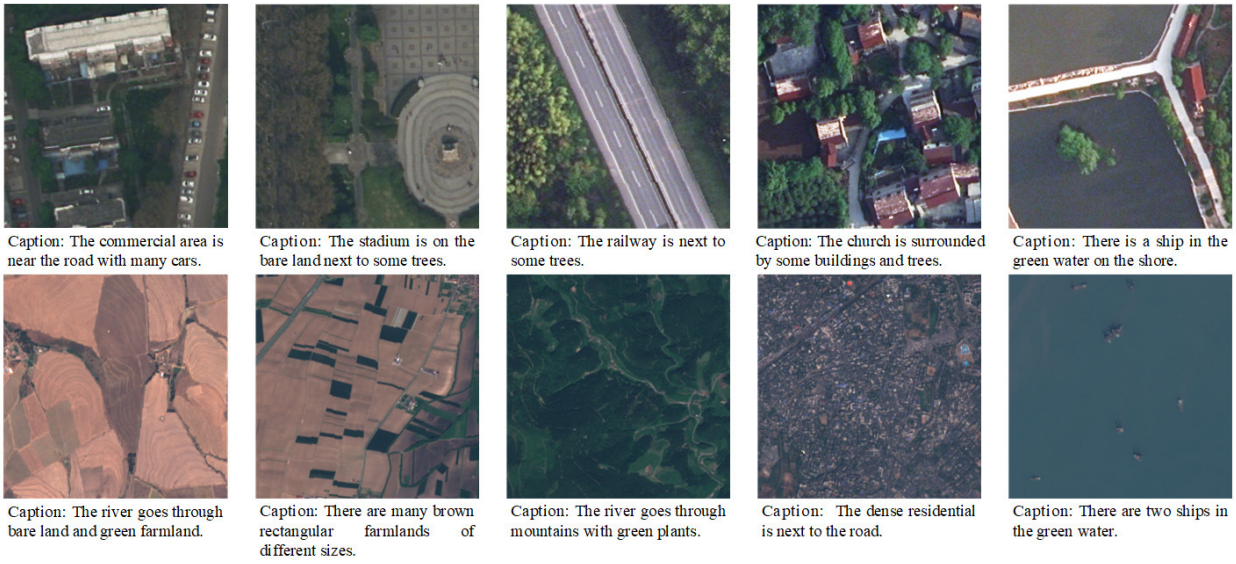



Fig. 7. Some caption examples on (top) Google Earth images and (bottom) Sentinel-2 images. The captions are generated by ASSMN methods.

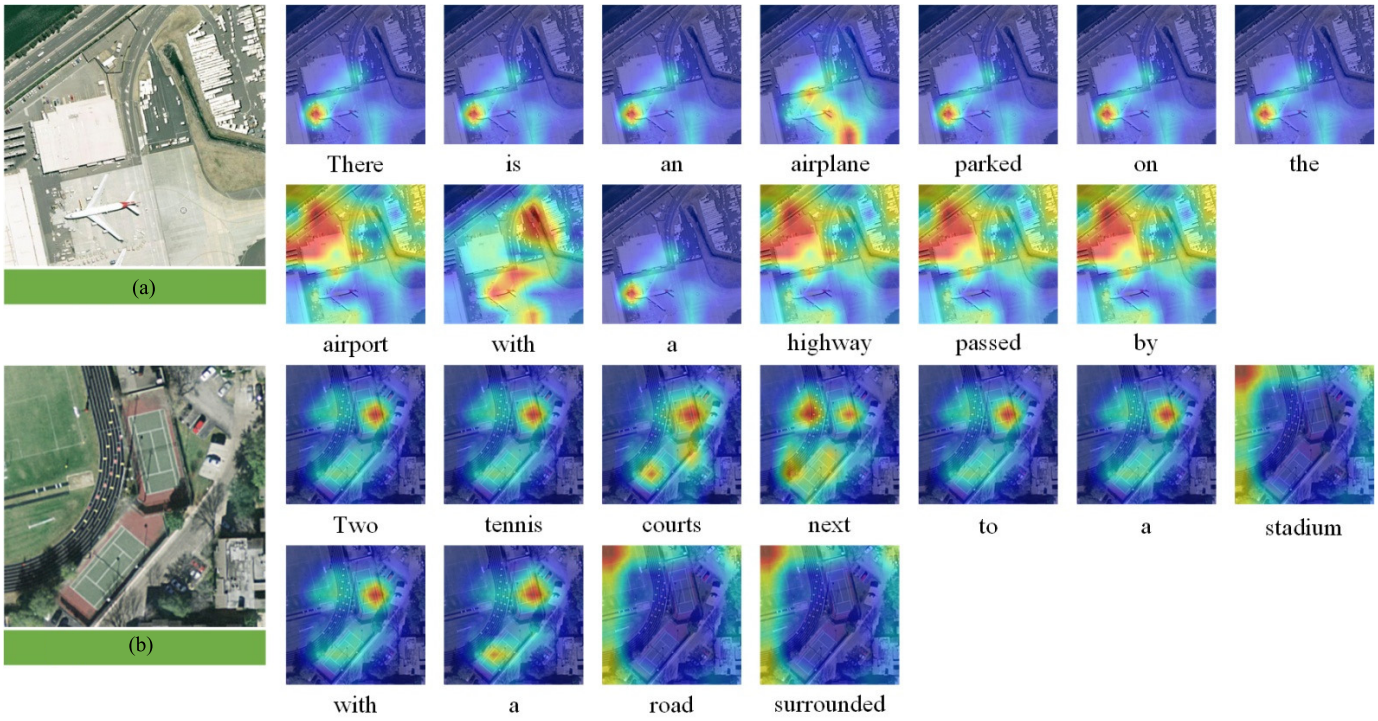



Fig. 8. Visualization of attended image regions along the caption generation processes for ASSMN. The red color indicates areas with high attention. (a) Airport. (b) Tennis courts.

ability. Furthermore, the NWPU dataset encompasses a broader range of object categories and scales, potentially enabling the trained model to capture richer remote sensing information. In conclusion, ASSMN can generate descriptions in most scenarios that are entirely novel to the model. While errors may still occur when describing weak or highly similar objects, our method demonstrates significant advantages in describing prominent objects and details. Based on the above analysis, proposed method exhibits a degree of transferability and holds practical application potential in real-world scenarios.

*3) Visualization of the Attention:* To more intuitively demonstrate the image regions focused on during sentence generation, the changes in the attention region during word generation are visualized in Fig. 8. Two remote sensing scenarios for ASSMN attention visualization are presented, such as an airport and a tennis court. While not all words precisely correspond to image locations, most activated regions align with words, such as aircraft at different scales, roads, and tennis courts within the images. Notably, certain consecutive words elicit responses to the same local features. For instance, in the airport scene, “parked on” and “highway

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passed by" cluster to two distinct local features, precisely reflecting the hierarchical structure of natural language from words to phrases to sentences.

### G. Limitation Analysis and Future Work

Building upon methods for remote sensing image captioning based on single visual encoders and hybrid visual encoders, we propose the ASSMN method. Within this framework, the MFE and SAA modules are constructed to capture comprehensive semantic information, fully accounting for multiscale objects and their attributes within remote sensing images. Furthermore, to fully leverage the relevant semantic information between remote sensing images, the SMR module is designed to store semantic information. This enables the model to effectively utilize relevant semantic information from other samples to generate accurate and complete captions. Despite the promising results achieved by ASSMN across multiple benchmark datasets, the proposed framework has certain limitations in handling remote sensing images containing small objects. Specifically, the model sometimes fails to accurately describe fine-grained objects that occupy only a small portion of the image. As shown in Fig. 7, the results displayed in the last column indicate that the model struggles to describe small plants in water bodies. This is because their style closely resembles that of the water itself and their area is very small. The current approach utilizes image semantic features extracted by a multiscale visual encoder for text description. It relies solely on cross-modal attention for interaction between visual and language components, lacking fine-grained alignment with language features. To address this limitation, future research may explore incorporating fine-grained alignment between images and text while avoiding excessive computational overhead, thereby enhancing the model's sensitivity to minute objects. Additionally, given that current models employ lightweight language decoders, while pretrained large language models demonstrate strong generalization and logical reasoning capabilities, future research could leverage large language models for generating captions for remote sensing images. This approach would endow the model with the rich world knowledge inherent in large language models, thereby better meeting the demands of real-world applications.

## V. Conclusion

In this article, we propose a novel remote sensing image captioning method, ASSMN, based on multiscale feature aggregation and global sample enhancement. Based on the CNN and transformer architectures, MFE, SAA, and SMR are introduced. To enhance the multiscale feature representation capability, we propose MFE to efficiently extract target features at different scales. Multilevel features are further aggregated by SAA to reduce the redundancy among visual features by aggregating the relevant feature vectors in the current visual feature mapping. To fully exploit the relevant information in the historical samples, feature storage is performed via SMR and injected into the model in a cross-attention manner, which ultimately results in robust and well-performing captions. Comparative analysis with state-of-the-art models shows that the model outperforms other models for captioning on three remote sensing image captioning datasets. In addition, the contribution of each module is analyzed through an ablation study. The generalizability of the proposed model was also validated through testing on Sentinel-2 and Google Earth images.

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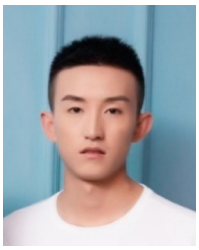
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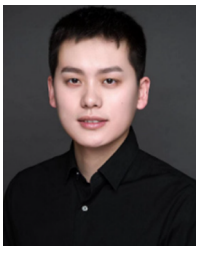
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